

Fejér–heat generators and Lipschitz control for the Weil quadratic functional

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Abstract

For a test function Φ in a suitable cone of even, compactly supported functions, the Weil quadratic functional $Q(\Phi)$ arises naturally from the Guinand–Weil explicit formula in analytic number theory. We establish three foundational results for the operator-theoretic study of this functional. First, we fix a normalization convention (T0) that identifies the Guinand–Weil frequency axis with the repository convention $\xi = \eta/(2\pi)$, ensuring compatibility with classical references. Second, we prove a cone density theorem (A1’): on any frequency compact $[-K, K]$, the cone generated by Fejér–heat windows $\Phi_{B,t}(\xi) = (1 - |\xi|/B)_+ \cdot e^{-4\pi^2 t \xi^2}$ is dense in the cone of continuous, even, nonnegative functions under the uniform norm. The approximation is explicit, with bandwidth $B = O(1/\varepsilon)$ and heat scale t depending on the mollification error. Third, we establish Lipschitz continuity (A2) of Q on each compact window: for $\Phi_1, \Phi_2 \in C_c([-K, K])$ even, one has $|Q(\Phi_1) - Q(\Phi_2)| \leq L_Q(K) \|\Phi_1 - \Phi_2\|_\infty$, where the Lipschitz constant $L_Q(K)$ is computed explicitly from the Archimedean density $a(\xi)$ and the finitely many prime weights in $[-K, K]$. The Fejér–heat tail outside the support contributes at most $t^{-1}e^{-t(\log N)^2}$, which is absorbed into the continuity budget. These results form the analytic foundation for the Toeplitz barrier and RKHS contraction arguments developed in [Part II]. This paper is the first of three developing an operator-theoretic approach to the Weil positivity criterion.

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1 Introduction

1.1 The Weil functional and prime distribution

Let Λ denote the von Mangoldt function, and for $n \geq 2$ define the prime nodes and Weil weights

$$\xi_n := \frac{\log n}{2\pi}, \quad w_Q(n) := \frac{2\Lambda(n)}{\sqrt{n}}. \quad (1)$$

For an even test function $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ supported on a compact interval $[-K, K]$, the *Weil quadratic functional* is

$$Q(\Phi) := \int_{-K}^K a^*(\xi) \Phi(\xi) d\xi - \sum_{\xi_n \in [-K, K]} w_Q(n) \Phi(\xi_n), \quad (2)$$

where

$$a^*(\xi) := 2\pi \left(\log \pi - \Re \psi \left(\frac{1}{4} + i\pi\xi \right) \right) \quad (3)$$

is the *Archimedean density* arising from the Guinand–Weil explicit formula [2, 1], and $\psi(z) = \Gamma'(z)/\Gamma(z)$ denotes the digamma function. The functional Q encodes information about the distribution of prime numbers through the Fourier-analytic structure of the explicit formula. Understanding when Q is nonnegative on natural test cones is a central problem in analytic number theory.

Our approach to this problem decomposes into a sequence of analytic modules:

Table 1: Analytic modules for the Weil positivity analysis.

Module	Statement	Reference
(T0)	Normalization crosswalk: Guinand–Weil $\leftrightarrow$ repository	This paper
(A1')	Fejér–heat cone dense in $C_{\text{even}}^+([-K, K])$	This paper
(A2)	Q Lipschitz on $C_c([-K, K])$	This paper
(A3)	Toeplitz barrier: $\inf_{\theta} P_A(\theta) \geq c_* > 0$	[Part II]
(RKHS)	Prime cap: $\ T_P\ \leq \rho(t) < c_*/4$	[Part II]

The synthesis of these modules appears in [Part III].

1.2 Main results

We now state the main results of this paper.

Theorem 1.1 (T0: Normalization crosswalk). *Under the Fourier convention $\widehat{\varphi}(\xi) = \int_{\mathbb{R}} \varphi(t) e^{-2\pi i t \xi} dt$, the repository quadratic functional $Q(\varphi)$ coincides with the classical Guinand–Weil functional $Q_{\text{GW}}(\varphi_{\text{GW}})$ under the change of variables $\eta = 2\pi\xi$, $\varphi_{\text{GW}}(\eta) = \varphi(\eta/(2\pi))$. The Archimedean density transforms as $a^*(\xi) = 2\pi a(\xi)$ where $a(\xi) = \log \pi - \Re \psi(\frac{1}{4} + i\pi\xi)$.*

The test functions we consider are *Fejér–heat windows*:

$$\Phi_{B,t}(\xi) := \left(1 - \frac{|\xi|}{B} \right)_+ \cdot e^{-4\pi^2 t \xi^2}, \quad (4)$$

where $(x)_+ := \max(x, 0)$, $B > 0$ is the bandwidth, and $t > 0$ is the heat scale. For $\tau \in [-K, K]$, the symmetrized atoms are

$$\Phi_{B,t,\tau}(\xi) := \frac{1}{2} [\Lambda_B(\xi - \tau)\rho_t(\xi - \tau) + \Lambda_B(\xi + \tau)\rho_t(\xi + \tau)], \quad (5)$$

where $\Lambda_B(x) = (1 - |x|/B)_+$ is the Fejér kernel (tent function) and $\rho_t(x) = (4\pi t)^{-1/2}e^{-x^2/(4t)}$ is the heat kernel.

Theorem 1.2 (A1': Cone density). *Let $K = [-R, R]$ with $R > 0$. The closed convex cone $\mathcal{C}$ generated by finite nonnegative combinations of the symmetrized atoms $\{\Phi_{B,t,\tau}\}$ with $\tau \in [-R, R]$ and $B \geq B_0(R, \varepsilon)$ is dense in $C_{\text{even}}^+([-R, R])$ under the uniform norm $\|\cdot\|_\infty$.*

The explicit dependence is $B_0 = O(R/\varepsilon)$; see Lemma 3.2 for the precise constant.

Theorem 1.3 (A2: Lipschitz continuity). *Fix $K = [-R, R]$. For even $\Phi_1, \Phi_2 \in C_c([-R, R])$, one has*

$$|Q(\Phi_1) - Q(\Phi_2)| \leq L_Q(K) \|\Phi_1 - \Phi_2\|_\infty, \quad (6)$$

where the Lipschitz constant is

$$L_Q(K) := \|a^*\|_{L^1(K)} + \sum_{\xi_n \in K} w_Q(n). \quad (7)$$

The sum is finite because only $n \leq e^{2\pi R}$ satisfy $\xi_n \in K$.

1.3 Proof overview

The proof of Theorem 1.1 is a direct calculation: the Jacobian $d\eta = 2\pi d\xi$ absorbs into the density a^* , and the evenness of φ ensures the prime sum matches under $\xi_n = (\log n)/(2\pi)$. We include it for completeness and to fix notation used throughout the trilogy.

The density theorem (A1') proceeds in three steps:

- (i) *Mollification*: Convolve with the heat kernel ρ_t to produce $g = \tilde{f} * \rho_t \in C^\infty$, with $\|g - f\|_\infty < \varepsilon/3$ for small t .
- (ii) *Positive Riemann sums*: Approximate g by a finite sum of heat translates with nonnegative coefficients $g(\tau_j^*)$.
- (iii) *Fejér truncation*: Replace the unbounded Gaussian by the compactly supported Fejér kernel Λ_B , with B large enough that $|\Lambda_B(\xi - \tau) - 1| < \varepsilon/3$ on $[-R, R]$.

The triangle inequality collects the errors. The symmetrization preserves evenness.

The Lipschitz bound (A2) follows from two observations: (i) the Archimedean density a^* is bounded on $[-R, R]$, hence the integral term is Lipschitz in Φ ; (ii) the prime sum is a finite sum of point evaluations, and finiteness is guaranteed by $\xi_n \in [-R, R] \Leftrightarrow n \leq e^{2\pi R}$.

1.4 Notation

We collect the notational conventions used throughout.

- **Prime nodes and weights**: $\xi_n = (\log n)/(2\pi)$, $w_Q(n) = 2\Lambda(n)/\sqrt{n}$.
- **Bandwidth window**: $W_K = [-K, K]$, the frequency compact.
- **Fejér kernel**: $\Lambda_B(\xi) = (1 - |\xi|/B)_+$, supported on $[-B, B]$.
- **Heat kernel**: $\rho_t(x) = (4\pi t)^{-1/2}e^{-x^2/(4t)}$, with $\int_{\mathbb{R}} \rho_t = 1$.
- **Archimedean density**: $a(\xi) = \log \pi - \Re \psi(\frac{1}{4} + i\pi\xi)$, $a^*(\xi) = 2\pi a(\xi)$.
- **Asymptotic notation**: $O(\cdot)$ denotes absolute constants; $O_K(\cdot)$ allows dependence on K .

2 Normalization: the Guinand–Weil crosswalk (T0)

2.1 Fourier conventions

We adopt the Fourier normalization

$$\widehat{\varphi}(\xi) = \int_{\mathbb{R}} \varphi(t) e^{-2\pi i t \xi} dt, \quad \varphi(t) = \int_{\mathbb{R}} \widehat{\varphi}(\xi) e^{2\pi i t \xi} d\xi, \quad (8)$$

using the Lebesgue measure $d\xi$ on the frequency side. All test functions are even, so the sine components vanish.

2.2 The Guinand–Weil functional

The classical Guinand–Weil functional [1, 2] is defined on even test functions φ_{GW} on the frequency axis $\eta \in \mathbb{R}$ by

$$Q_{\text{GW}}(\varphi_{\text{GW}}) := \int_{\mathbb{R}} \left(\log \pi - \Re \psi \left(\frac{1}{4} + \frac{i\eta}{2} \right) \right) \varphi_{\text{GW}}(\eta) d\eta - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} [\varphi_{\text{GW}}(\log n) + \varphi_{\text{GW}}(-\log n)]. \quad (9)$$

Proof of Theorem 1.1. Set $\eta = 2\pi\xi$ and $\varphi(\xi) = \varphi_{\text{GW}}(2\pi\xi)$. For the Archimedean integral:

$$\int_{\mathbb{R}} \left(\log \pi - \Re \psi \left(\frac{1}{4} + \frac{i\eta}{2} \right) \right) \varphi_{\text{GW}}(\eta) d\eta = \int_{\mathbb{R}} 2\pi \left(\log \pi - \Re \psi \left(\frac{1}{4} + i\pi\xi \right) \right) \varphi(\xi) d\xi,$$

using $d\eta = 2\pi d\xi$ and $\psi(\frac{1}{4} + \frac{i\eta}{2}) = \psi(\frac{1}{4} + i\pi\xi)$.

For the prime sum, $\varphi_{\text{GW}}(\pm \log n) = \varphi(\pm \xi_n)$ with $\xi_n = (\log n)/(2\pi)$. Since φ is even:

$$\varphi_{\text{GW}}(\log n) + \varphi_{\text{GW}}(-\log n) = \varphi(\xi_n) + \varphi(-\xi_n) = 2\varphi(\xi_n).$$

Thus

$$\sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} [\varphi_{\text{GW}}(\log n) + \varphi_{\text{GW}}(-\log n)] = \sum_{n \geq 2} \frac{2\Lambda(n)}{\sqrt{n}} \varphi(\xi_n) = \sum_{n \geq 2} w_Q(n) \varphi(\xi_n).$$

Combining, $Q(\varphi) = Q_{\text{GW}}(\varphi_{\text{GW}})$ as claimed. $\square$

Remark 2.1. The doubling convention $w_Q(n) = 2\Lambda(n)/\sqrt{n}$ for nodes at $\xi_n > 0$ is equivalent to placing unit weights at both $\pm \xi_n$; evenness makes the two conventions identical.

3 Cone density on frequency compacts (A1')

We establish that the Fejér–heat cone is dense in the cone of continuous, even, nonnegative functions on any compact.

3.1 Heat mollification

The heat kernel $\rho_t(x) = (4\pi t)^{-1/2} e^{-x^2/(4t)}$ satisfies:

- (a) $\rho_t \geq 0$ and $\int_{\mathbb{R}} \rho_t = 1$ (probability density);
- (b) $\rho_t \rightarrow \delta_0$ as $t \rightarrow 0^+$ in the sense of distributions;
- (c) For $f \in C_b(\mathbb{R})$, $\|f * \rho_t - f\|_{\infty} \rightarrow 0$ as $t \rightarrow 0^+$ (uniform approximation).

Lemma 3.1 (Mollification error). *Let $f \in C_c(\mathbb{R})$ with $\text{supp } f \subseteq [-L, L]$. For any $\varepsilon > 0$, there exists $t_0 = t_0(f, \varepsilon) > 0$ such that for all $t \in (0, t_0]$:*

$$\|f * \rho_t - f\|_\infty < \varepsilon.$$

*Moreover, if $f \geq 0$ and f is even, then $f * \rho_t \geq 0$ and $f * \rho_t$ is even.*

Proof. Standard properties of approximate identities. Nonnegativity follows from $f \geq 0$ and $\rho_t \geq 0$. Evenness: if $f(x) = f(-x)$ and $\rho_t(x) = \rho_t(-x)$, then $(f * \rho_t)(-x) = (f * \rho_t)(x)$. $\square$

3.2 Fejér truncation

Lemma 3.2 (Fejér approximation). *For $|\xi|, |\tau| \leq R$ and $B > 0$:*

$$|\Lambda_B(\xi - \tau) - 1| \leq \frac{|\xi - \tau|}{B} \leq \frac{2R}{B}.$$

In particular, choosing $B \geq 6R/\varepsilon$ ensures $|\Lambda_B(\xi - \tau) - 1| < \varepsilon/3$ on $[-R, R]^2$.

Proof. For $|x| \leq B$, $\Lambda_B(x) = 1 - |x|/B$, so $|\Lambda_B(x) - 1| = |x|/B$. Since $|\xi - \tau| \leq |\xi| + |\tau| \leq 2R$, the bound follows. $\square$

3.3 Proof of cone density

Proof of Theorem 1.2. Fix $f \in C_{\text{even}}^+([-R, R])$ and $\varepsilon > 0$. Extend f by zero to $\tilde{f} \in C_c(\mathbb{R})$.

Step 1 (Mollification). By Lemma 3.1, there exists $t > 0$ such that $g := \tilde{f} * \rho_t$ satisfies

$$\sup_{|\xi| \leq R} |g(\xi) - f(\xi)| < \varepsilon/3.$$

The function g is nonnegative, even, and smooth.

Step 2 (Riemann sums). Choose a uniform partition $-R = \tau_0 < \tau_1 < \dots < \tau_N = R$ with mesh Δ small enough that

$$g_R(\xi) := \sum_{j=0}^{N-1} g(\tau_j^*) \rho_t(\xi - \tau_j^*) \Delta$$

satisfies $\|g_R - g\|_{L^\infty([-R, R])} < \varepsilon/3$. The coefficients $g(\tau_j^*) \cdot \Delta$ are nonnegative.

Step 3 (Fejér truncation). Choose $B \geq 6R/\varepsilon$ per Lemma 3.2. Define

$$h(\xi) := \sum_{j=0}^{N-1} g(\tau_j^*) \Delta \Phi_{B,t,\tau_j^*}(\xi).$$

For $|\xi| \leq R$, $|\Lambda_B(\xi \pm \tau_j^*) - 1| < \varepsilon/3$, so

$$|h(\xi) - g_R^{\text{sym}}(\xi)| \leq \frac{\varepsilon}{3} \sum_j g(\tau_j^*) \Delta \leq C_f \cdot \frac{\varepsilon}{3},$$

where $C_f = \int_{-R}^R g(\tau) d\tau$. Rescaling ε by $3 \max(1, C_f)$:

$$\|h - g_R^{\text{sym}}\|_\infty < \varepsilon/3.$$

Step 4 (Triangle inequality). Since g is even, g_R^{sym} satisfies $\|g_R^{\text{sym}} - g\|_\infty < \varepsilon/3$. Combining:

$$\|h - f\|_\infty \leq \|h - g_R^{\text{sym}}\|_\infty + \|g_R^{\text{sym}} - g\|_\infty + \|g - f\|_\infty < \varepsilon.$$

Since h is a finite nonnegative combination of atoms Φ_{B,t,τ_j^*} , this completes the proof. $\square$

Remark 3.3 (Parameter scaling). The bandwidth B and heat scale t depend on the compact $[-R, R]$ and accuracy ε . No uniform bound $\sup_K B(K)$ is required for density; the schedules $K \mapsto B(K)$, $K \mapsto t(K)$ may grow with K .

4 Lipschitz continuity of Q (A2)

4.1 Finiteness of the prime sampler

Lemma 4.1 (Local finiteness). *Fix $K > 0$. For $\Phi \in C_c([-K, K])$ even, the prime sum*

$$\sum_{n \geq 2} w_Q(n) \Phi(\xi_n)$$

is a finite sum: only $n \leq e^{2\pi K}$ contribute.

Proof. $\Phi(\xi_n) = 0$ for $|\xi_n| > K$, i.e., for $(\log n)/(2\pi) > K$, equivalently $n > e^{2\pi K}$. □

4.2 The Lipschitz bound

Proof of Theorem 1.3. Write $Q(\Phi_1) - Q(\Phi_2) = I_A + I_P$ where

$$I_A = \int_{-R}^R a^*(\xi) (\Phi_1 - \Phi_2)(\xi) d\xi, \quad I_P = - \sum_{\xi_n \in [-R, R]} w_Q(n) (\Phi_1 - \Phi_2)(\xi_n).$$

Archimedean term. Since a^* is continuous on $[-R, R]$:

$$|I_A| \leq \int_{-R}^R |a^*(\xi)| |\Phi_1 - \Phi_2|(\xi) d\xi \leq \|a^*\|_{L^1(K)} \|\Phi_1 - \Phi_2\|_\infty.$$

Prime term. By Lemma 4.1, the sum is finite:

$$|I_P| \leq \sum_{\xi_n \in K} w_Q(n) |\Phi_1(\xi_n) - \Phi_2(\xi_n)| \leq \left(\sum_{\xi_n \in K} w_Q(n) \right) \|\Phi_1 - \Phi_2\|_\infty.$$

Adding the bounds gives (6) with $L_Q(K) = \|a^*\|_{L^1(K)} + \sum_{\xi_n \in K} w_Q(n)$. □

4.3 Tail control for Fejér–heat windows

Lemma 4.2 (Gaussian tail bound). *For $\Phi_{B,t}$ a Fejér–heat window and $N \geq e^2$:*

$$\text{Tail}(t; N) := \sum_{n > N} w_Q(n) \Phi_{B,t}(\xi_n) \leq \frac{1}{t} e^{-t(\log N)^2}.$$

Proof. For the heat factor, $e^{-4\pi^2 t \xi_n^2} = e^{-t(\log n)^2}$. Thus

$$\sum_{n > N} w_Q(n) \Phi(\xi_n) \leq \sum_{n > N} \frac{2 \log n}{\sqrt{n}} e^{-t(\log n)^2}.$$

Via the integral comparison $\sum_{n > N} \leq \int_{N-1}^\infty$ and the substitution $y = \log x$:

$$\int_{\log N}^\infty y e^{-ty^2} dy = \frac{1}{2t} e^{-t(\log N)^2}.$$

Multiplying by 2 gives the stated bound. □

Remark 4.3 (Leakage control). When Fejér–heat windows are used on $[-K, K]$, the Gaussian factor produces exponentially small leakage outside the compact. This tail is absorbed into the (A2) continuity budget.

5 Discussion

5.1 Summary

We have established three foundational results:

1. **T0 (Normalization):** The repository convention $\xi = \eta/(2\pi)$ matches Guinand–Weil with Jacobian factor $a^* = 2\pi a$.
2. **A1' (Density):** Fejér–heat atoms generate a cone dense in $C_{\text{even}}^+([-K, K])$.
3. **A2 (Lipschitz):** Q is Lipschitz with explicit constant $L_Q(K)$ on each compact.

5.2 Relation to Part II

The density (A1') and continuity (A2) results reduce the positivity problem $Q \geq 0$ on the full Weil cone to verifying $Q \geq 0$ on Fejér–heat atoms. In [Part II], we establish:

- **A3 (Toeplitz barrier):** The Archimedean symbol $P_A(\theta)$ satisfies $\inf_{\theta} P_A(\theta) \geq c_* = 11/10$.
- **RKHS cap:** The prime operator T_P satisfies $\|T_P\|_{\mathcal{H}_t \rightarrow \mathcal{H}_t} < c_*/4$.

Together, these imply $Q(\Phi) \geq (c_* - c_*/4)\|\Phi\|^2 > 0$ for atoms.

5.3 Open questions

1. Can the Lipschitz constant $L_Q(K)$ be computed in closed form for small K ?
2. What is the optimal heat scale t minimizing the total approximation error in (A1')?
3. Are there other test cones (beyond Fejér–heat) with similar density and continuity properties?

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